

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 3282	Course Title: Data Structures using C
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- Explain the fundamental concept of Data Structures and Algorithms.
- Emphasize the importance of data structures in designing and implementing efficient algorithms.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Control Structures, Arrays, Functions		Problem Solving and Programming	II

Course Outcomes :

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate fundamental concepts of Data Structures and Algorithms.	10	Understanding Applying
CO2	Construct different Searching and Sorting Algorithms	10	Applying
CO3	Develop different operations of Stacks, Queues and Linked Lists	13	Applying
CO4	Demonstrate nonlinear Data Structures like Tree and Graph	10	Understanding Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	1					
CO2	3	1					
CO3	3	1					
CO4	3	1					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate fundamental concepts of Data Structures and Algorithms.		
M1.01	Explain the concept of Pointers, Structures and Dynamic Memory Allocation	3	Understanding
M1.02	Develop the concept of Files with example	3	Applying
M1.03	Describe data structure and their classification	3	Understanding
M1.04	Define Asymptotic Analysis of Algorithms- The “Big-Oh” Notation, Big Omega Notation, Recursive Algorithms	1	Remembering
Contents: Advanced C concepts: Concept of Pointers, Files, Dynamic Memory Allocation. Data Structures: Definition, Classification- Linear Data Structures, Non-Linear Data Structures Algorithms: Definition, Asymptotic Analysis of Algorithms- The “Big-Oh” Notation, Big Omega Notation, Recursive Algorithms			
CO2	Construct different Searching and Sorting Algorithms		
M2.01	Develop Linear Search and Binary Search	3	Applying
M2.02	Demonstrate Bubble Sort and Selection Sort with example	3	Applying
M2.03	Demonstrate Merge Sort with example	2	Applying
M2.04	Demonstrate Quick Sort with example	2	Applying
Contents: Searching Algorithms: Linear Search, Binary Search.			

Sorting Algorithms: Bubble Sort, Selection Sort, Quick Sort, Merge Sort			
CO3	Develop different operations of Stacks, Queues and Linked Lists		
M3.01	Explain stack creation and operations with examples	2	Understanding
M3.02	Apply stack operations to Convert infix expression to postfix expression and evaluate postfix expression.	2	Applying
M3.03	Explain queue creation and operations with examples	2	Understanding
M3.05	Explain circular queue and Dequeue operations with examples	2	Understanding
M3.06	Outline the concept of Linked List and its representation in memory.	1	Understanding
M3.07	Implement a Singly Linked List and Doubly Linked List.	2	Applying
M3.08	Demonstrate Linked List Representation of Stack and Queue	2	Applying
Contents: Stacks: Introduction to Stacks - Array Representation of Stacks - Operations on a Stack - Applications of Stacks - Infix-to-Postfix Conversion - Evaluating Postfix Expressions. Queues: Introduction to Queues - Array Representation of Queues - Operations on a Queue - Types of Queues: Circular Queue – Dequeue. Linked Lists: Singly Linked List - Representation in Memory, Static, Dynamic - Operations on a Singly Linked List – Insertion, Deletion, Searching and Traversal. Types of Linked List- Circular Linked Lists, Doubly Linked Lists, Linked List Representation of Stack and Queue			
CO4	Demonstrate nonlinear Data Structures like Tree and graph		
M4.01	Explain the essential key terms related to Graphs and different classification of graphs.	1	Understanding
M4.02	Illustrate the different representations of a Graph.	2	Applying
M4.03	Demonstrate Graph Traversal algorithms like BFS and DFS.	1	Applying
M4.04	Explain the essential key terms related to Trees.	1	Understanding
M4.05	Show the representation of a Binary Tree	1	Applying
M4.06	Construct different Tree Traversal Algorithms like Inorder, Preorder and Postorder	2	Applying
M4.07	Demonstrate a Binary Search Tree with different operations like Creation, Insertion, Search and Deletion.	2	Applying

Contents:

Graphs: Graph Terminologies - Vertex, Edge, Adjacent vertices, Self-loop, Parallel edges, Isolated vertex, Degree of vertex, Pendant vertex, Subgraph, Paths and Cycles.

Types of Graphs - Directed, Undirected, Simple, Complete, Cyclic, Acyclic, Bipartite, Complete Bipartite, Connected, Disconnected and Regular. Representation of Graphs ,Graph Traversals-BFS, DFS

Trees: Definition - Basic Terminologies - Node, Parent, Child, Link, Root, Leaf, Level, Height of a tree and node, Depth of a tree and node, Degree of a tree and node, sibling, Ancestors, Path, Path Length.

Binary Tree- Types of Binary Trees: Full, Complete.

Representations of a Binary Tree using Linked Lists

Tree Traversal Algorithms- Inorder, Preorder, Postorder.

Operations on a Binary Search Tree: Insertion – Traversal – Deletion– Count number of nodes – Height of a BST

Text / Reference

T/R	Author/Book Title
T1	Data Structures using C, E. Balagurusamy, Tata Mcgraw Hill
T2	Programming with ANSI & Turbo C, Ashok Kamthane, Second Edition, Pearson Education.
R1	Let us C, Yashavant P Kanetkar, 14 th Edition, BPB publication, ISBN 9788183331630
R2	Dharmender Singh Kushwaha, Arun Kumar Misra.
R3	Programming in C and Data Structure, P.B.Kotur, Sapna Book house
R4	Fundamentals of Data Structures in C++, E.Horowitz- S.Sahni, Galgotia-2006
R5	Data Structures and Algorithm Analysis in C++, M.A.Weiss, Pearson Education- Fourth Edition
R6	Data Structures, Algorithms and Applications in C++, Sartaj Sahni, University Press
R7	Data Structures using C and C++ by Yedidyah Langsam, Moshe J. Augenstein and Aron M.Tananbaum, PHI.2002

Online Resources

Sl No	Website Link
1	https://www.cet.edu.in/noticefiles/280_DS%20Complete.pdf
2	http://spoken-tutorial.org/tutorial-search/?search_foss=C++andC++&search_language=English
3	http://www.tutorialspoint.com/cprogramming/
4	http://www.indiabix.com/online-test/c-programming-test